Dig Dis Sci (2007) 52:1313–1325
DOI 10.1007/s10620-006-9107-3

ORIGINAL PAPER

Asymptomatic Cholelithiasis: Is Cholecystectomy Really Needed? A Critical Reappraisal 15 Years After the Introduction of Laparoscopic Cholecystectomy

George H. Sakorafas · Dimitrios Milingos · George Peros

Received: 13 December 2005 / Accepted: 12 January 2006 / Published online: 28 March 2007
© Springer Science+Business Media, Inc. 2007

Abstract Asymptomatic cholelithiasis is increasingly diagnosed today, mainly as a result of the widespread use of abdominal ultrasonography for the evaluation of patients for unrelated or vague abdominal complaints. About 10–20% of people in most western countries have gallstones, and among them 50–70% are asymptomatic at the time of diagnosis. Asymptomatic gallstone disease has a benign natural course; the progression of asymptomatic to symptomatic disease is relatively low, ranging from 10–25%. The majority of patients rarely develop gallstone-related complications without first having at least one episode of biliary pain ("colic"). In the prelaparoscopy era, (open) cholecystectomy was generally performed for symptomatic disease. The minimally invasive laparoscopic cholecystectomy refueled the discussion about the optimal management of asymptomatic cholelithiasis. Despite some controversy, most authors agree that the vast majority of subjects should be managed by observation alone (expectant management). Selective cholecystectomy is indicated in defined subgroups of subjects, with an increased risk for the development of gallstone-related symptoms and complications. Concomitant cholecystectomy is a reasonable option for good-risk patients with asymptomatic cholelithiasis undergoing abdominal surgery for unrelated conditions. Routine cholecystectomy for all subjects with silent gallstones is a too aggressive management option, not indicated for most subjects with asymptomatic cholelithiasis. An in-depth knowledge of the natural history of gallstone disease is required to select the optimal management option for the individual subject with silent gallstones. Management options should be extensively discussed with the patient; he or she should be actively involved in the process of therapeutic decision making.

Keywords Silent gallstones · Laparoscopic cholecystectomy · Porcelain gallbladder · Incidental cholecystectomy · Concomitant cholecystectomy · Gallbladder carcinoma · Cancer

G. H. Sakorafas (✉) · D. Milingos · G. Peros
4th Department of Surgery, Athens University, Medical School, "ATTIKON" University Hospital, Arkadias 19–21, GR-115 26 Athens, Greece
e-mail: georgesakorafas@yahoo.com

Introduction

Cholelithiasis constitutes one of the most common causes of hospitalization due to gastrointestinal problems in developed countries and accounts for an important part of health care expenditure. It shows a worldwide distribution and a high incidence. especially among particular nations and geographic areas in the world. The definite treatment of choice for gallstone disease is cholecystectomy. Until the introduction of laparoscopy, surgery (open cholecystectomy) was generally indicated only for patients with symptomatic and complicated disease [1, 2]. However the establishment of laparoscopic cholecystectomy as the gold standard for the treatment of cholelithiasis has created new interest for the management of patients with silent gallstones. As a minimally invasive technique, with many well-known advantages over open cholecystectomy, laparoscopic cholecystectomy renewed the interest and fueled the discussion regarding the optimal management of asymptomatic gallstone disease. Today, asymptomatic cholelithiasis is a very common clinical entity; the surgeon sees an increasing number of such subjects, owing to the widespread availability and use of abdominal ultrasonography for the investigation of a variety of abdominal diseases or vague abdominal complaints and in cases of routine checkup. Management of

such incidentally discovered gallstones poses a real dilemma for both the physician and the patient; conclusive evidence of benefits of cholecystectomy is lacking [3].

The aim of the present study was to critically discuss and summarize currently available data regarding optimal management of asymptomatic patients, taking into account the natural history of asymptomatic cholelithiasis, the potential for the development of complicated gallstone disease, and the potential morbidity and mortality of cholecystectomy.

Asymptomatic cholelithiasis: the extent of the problem

When discussing asymptomatic cholelithiasis, the first priority is to define this entity. *Asymptomatic cholelithiasis* exists when gallstones are detected in the absence of gallstone-related symptoms, such as history of biliary pain (pain in the epigastrium or right upper abdominal quadrant that may radiate to the patient's back or to the right scapula), or gallstone-related complications such as acute cholecystitis, cholangitis, or pancreatitis (Rome Group for the Epidemiology and Prevention of Cholelithiasis [GREPCO]) [4]. Other nonspecific symptoms or vague dyspeptic problems, such as epigastric discomfort, dyspepsia, flatulence, nausea, abdominal gurgling noises, or pain outside the right hypochodrium, cannot be considered as symptomatic cholelithiasis and could easily be attributed to other gastrointestinal diseases (such as peptic ulcer disease, irritable bowel syndrome) [5, 6]. Therefore, and when discussing surgery under this situation, the patient should understand that these symptoms may persist following cholecystectomy. However, the fact that patients with that kind of nonspecific symptomatology may sometimes benefit from cholecystectomy could indicate that cholelithiasis contributes, at least to some extent, to that vague clinical picture (see Selective Cholecystectomy—Vague Dyspeptic Symptoms) [7].

In 1992, it was estimated that 10–15% of the adult population in the USA had gallstones (which amounted to more than 20 million people), with women suffering twice as much as men [4, 5, 8–10]. About one million patients are newly diagnosed annually (10) and among them 50–70% are asymptomatic at diagnosis (6, 8). Approximately 600,000 patients underwent cholecystectomy in 1991 in the USA [11] and about 4–7% of those operated had asymptomatic cholelithiasis [12]. The annual cost for the treatment of those patients had been estimated to be around $5 billion (estimates from 1993) [10].

Similar results have been reported in Europe. The Rome Group for the Epidemiology and Prevention of Cholelithiasis found gallstones in 8% of Roman male civil servants between the age of 20 and 69 years. Fewer than 8% had a history of symptoms compatible with biliary colic [13, 14]. The same group found a prevalence of as high as 25% in female civil servants in the 60- to 64-year age group [13, 14]. One third reported at least one episode of biliary pain over a period of 5 years [4]. In the Italian Multicenter Italian Study on Cholelithiasis, 29,739 participants were examined by ultrasound and questionnaire with respect to the presence of gallstones and related symptoms [15]. The prevalence of gallstones for women was 10.5% and for men 6.5%. This increased to 18.9% and 9.5%, respectively, when subjects who had already undergone cholecystectomy were added. A linear increase in prevalence was noted with age in both genders. The vast majority of subjects with gallstones were asymptomatic (84.9% of women and 87% of men). Similar numbers were found in a Scandinavian study, where the overall prevalence of gallstones was 15%, with women having a higher prevalence both at 40 and at 60 years of age as compared with men (11% and 25% versus 4% and 15%, respectively) [16]. In the United Kingdom, around 5,500,000 people have gallstones and more than 50,000 cholecystectomies are performed each year [17].

It therefore appears that the overall prevalence of gallstone disease in industrialized countries is between 10% and 20%. The prevalence rises with age in both genders (close to 10 per 1,000 subjects per year) [18]. At the age of 65, about 30% of women have gallstones, and by the age of 80, 60% of both men and women have them [19]. These data confirm the high prevalence of gallstone disease and also shows that most patients are unaware of it.

Natural history of asymptomatic gallstones

> There is no innocent gallstone
>
> —William J. Mayo, MD, 1904
>
> It is unfortunate that so few appreciate from what small causes diseases come.
>
> —Charles H. Mayo, MD, 1902

About 100 years after the above-noted Mayo dictum about gallstones, there is now enough evidence that most incidentally discovered, clinically silent gallstones rarely have clinical significance [3, 20–24]. In most Western countries, the majority of patients with asymptomatic cholelithiasis remain asymptomatic throughout their life, and do not require any treatment [25]; in fact, these subjects live and die with their gallstones having never caused pain or other medical problems. Autopsy studies showed that more than 90% of autopsied patients with gallstone disease died from unrelated causes. Death as the ultimate complication from asymptomatic gallstones is very rare (~3–7% of deaths), usually in the elderly as a consequence of biliary or postoperative complications [26–28].

Table 1 Natural history of gallstones

Author	Patients (*n*)	Years	Biliary pain (%)	Biliary complications (%)
Gracie *et al*. [32]	123	15	18	2
Attili *et al*. [31]	118	10	26	3
Juhasz *et al*. [122]	110	6	22	
Wada *et al*. [35]	680	13	20	
McSherry *et al*. [14, 36]	135	4	10	0
Friedman *et al*. [38]	123	5		6
Thistle *et al*. [45]	193	2	31	
Cucchiaro *et al*. [37]	125	5	12	2
Angelico *et al*. [112]	426	10	38	
Zubler *et al*. [29]	32	5	25	3

Note. Modified from Vauthey and Saldinger [11].

According to the 1992 NIH Consensus Conference report [10], 10% of patients develop symptoms during the first 5 years after diagnosis and 20% by 20 years. Similar findings were reported by Zubler *et al*. [29], who found that only 10-18% of asymptomatic patients ever become symptomatic; the annual risk for developing biliary pain (misnamed as *biliary colic*) is 1–4% [17, 30] (Table 1). One could extrapolate that after 20 years, approximately two thirds of patients will remain symptom free [25]. These rates are in sharp contrast with those in symptomatic cholelithiasis, where the annual rates of developing complications and biliary pain are 1.2% and 50%, respectively [17]. According to the Italian GREPCO study, the annual complication rate of initially asymptomatic patients is 0.3–1.2% [31]. In this study, 151 subjects identified to have gallstones during the GREPCO–1984 study [4] were followed over a period of 10 years. At the beginning of the study, 118 patients were asymptomatic. The cumulative probability of developing complications after 10 years was 3% in the initially asymptomatic group and 7% in the symptomatic group.

In the significant study by Gracie and Ransohoff [32], 123 Michigan University faculty members (110 men and 13 women) found to have gallstones through routine screening were followed for 15 years. At 5, 10, and 15 years of follow-up, 10%, 15%, and 18%, respectively, became symptomatic. The approximate rate at which the subjects developed biliary pain was 2% per year for the first 5 years without a subsequent decrease over time. Three patients in this study developed biliary complications, all of which were preceded by biliary colic. Based on these results, the authors concluded that prophylactic cholecystectomy for asymptomatic cholelithiasis is not justified.

A longitudinal follow-up study of asymptomatic gallstones showed that over a 20-year period only 18% of patients developed biliary pain and that the mean yearly probability of the development of biliary pain is 2% during the first 5 years, 1% during the second 5 years, 0.5% during the third 5 years, and 0% during the fourth 5 years. None of these individuals died because of gallstone disease [33].

According to Hermann [34], 40–60% of persons with cholelithiasis remain asymptomatic, 60–70% present with mild symptoms of chronic cholecystitis (among them, 20% have manifestations that are difficult to interpret), 20% develop acute cholecystitis, and 10% develop complicated acute cholecystitis (i.e., cholecystitis accompanied by jaundice, cholangitis, or pancreatitis). Hermann concluded that the longer patients live with gallstones, the more likely they are to experience pain or complications [34].

In Japan, Wada and Imamura [35] found that 20% of patients with asymptomatic cholelithiasis turned symptomatic after a median follow-up of 13 years. Patients over the age of 70 were more likely to become symptomatic than patients under 70. McSherry *et al*. [36] followed 135 asymptomatic men and women with gallstones who were subscribers to the Health Insurance Plan of Greater New York. Ten percent developed symptoms and only 7% required cholecystectomy over a median follow-up of 46.3 months. Cucchiaro *et al*. [37] followed 125 asymptomatic patients for a period of 5 years. Fifteen patients developed symptoms during that time and two underwent emergency surgery for gallstone complications. Fifty-four patients died during that period because of unrelated causes (malignancies, cardiovascular disease, renal insufficiency).

Friedman *et al*. [38] observed 123 asymptomatic patients over 25 years in a prepaid health plan; serious or mild events (acute cholecystitis, acute biliary pancreatitis, obstructive jaundice) occurred in 4% of these asymptomatic patients. Wacha and Ungeheuer in their review [39] reported higher rates of conversion from asymptomatic to symptomatic state; according to these authors, as many as 50% of individuals with asymptomatic gallstones will be operated on or develop symptoms within 10–20 years after the initial diagnosis.

Aging is found to be associated with the development of symptoms or complications, and in particular if the follow-up is long enough, in an increasing percentage of patients, up to 30–50% [34, 36, 40, 41]. The incidence of choledocholithiasis at the time of cholecystectomy is directly related to age (9% in those 31–40 years old to 96% in those

80–90 years old) [5]. As is well known, choledocholithiasis may be associated with potentially serious complications, such as acute cholangitis and/or pancreatitis. This is of particular practical importance for the patient and the physician, because patients with advanced age exhibit higher morbidity and mortality rates, whereas the management of complicated gallstone disease may require more complicated procedures than simple laparoscopic cholecystectomy. However, this issue remains controversial; other authors observed that the probability of developing symptoms and/or complications fell steadily over time [10, 25, 31–33]; according to these investigators, the longer patients were asymptomatic, the less likely they were to develop symptoms.

In summary, most studies (conducted mainly in the 1980s) indicate that the progression of asymptomatic to symptomatic disease is relatively low, ranging from 10–25% [42–45]. The major concern when discussing the natural history of asymptomatic cholelithiasis is the possible development of a severe, potentially life-threatening complication, such as severe (necrotizing) pancreatitis or acute suppurative cholangitis. Existing data show, however, that the majority of patients rarely develop complications without first having at least one episode of biliary colic pain [5, 10, 17, 32, 36]; biliary colic usually occurs within the first 5 years of the initial diagnosis [20]. From a practical point of view, it would be very important for both the patient and the physician if we could recognize the subgroup of asymptomatic patients who will become symptomatic. Unfortunately, it is impossible, using local (such as number, size, nature, alteration in wall thickness or gallbladder contractility) or general factors (such as age, gender, associated comorbidities) to predict who—among asymptomatic patients—will ever develop symptoms or complications and when [3]. Some authors tried to classify asymptomatic patients into two groups: a low-risk group and a high-risk group. The low-risk patients are those with a functioning gallbladder whose calculi are >3 mm but <2 cm in diameter and radiolucent, and who are free of concomitant serious disease [5]. The high-risk patients are those more likely to develop acute cholecystitis, acute pancreatitis, or other complications of cholelithiasis and include those with large stones (>2.5 cm) and those with small multiple calculi (microlithiasis, stones <3 mm in diameter), biliary sludge, or both, who tend to develop acute cholangitis or pancreatitis [46–48]. The risk of complicated cholelithiasis is admittedly higher when the cystic duct is chronically obliterated [5].

Management of asymptomatic cholelithiasis

Treatment options for asymptomatic cholelithiasis include expectant management (observation alone) and cholecystectomy (laparoscopic), which can be performed either selectively (for selected subgroups of patients with asymptomatic cholelithiasis), routinely (for all patients with asymptomatic cholelithiasis), or concomitantly during another intraabdominal operation for an unrelated pathologic condition (e.g., cancer of the colon).

Expectant management (observation alone)

Because the majority of patients with asymptomatic cholelithiasis remain asymptomatic throughout their life, most authors agree that expectant management (watchful waiting) is the most reasonable treatment for the majority of these patients [3, 17, 20–23, 49–51). This approach avoids overtreatment (an unnecessary surgical procedure under general anesthesia) in the vast majority of asymptomatic patients who will never develop symptoms. The disadvantage of this approach is that no one can guarantee the patient that he or she will never suffer a potentially severe or even lethal complication of gallstone disease, such as gallbladder cancer or severe acute pancreatitis, usually at a more advanced age (Table 2). In this case, emergency surgery for a serious complication of cholelithiasis may be needed at a later date. Moreover, in such a setting, the operation is much more complicated and it is performed more frequently by laparotomy, which may increase morbidity and mortality, especially in older patients with comorbidities [5]. However, the patient should recognize that this theoretical possibility is rare. Usually the conversion from the asymptomatic to symptomatic state happens by the occurrence of a biliary colic, thereby indicating the need for surgery in the—*then symptomatic*—patient. In conclusion, and considering that the vast majority of asymptomatic patients remain asymptomatic throughout their life and that most asymptomatic patients develop symptoms before the occurrence of complications, prophylactic surgical therapy is not justified—with a few exceptions (see Selective Cholecystectomy). According to the NIH Consensus Conference report [10] "the availability of laparoscopic cholecystectomy should not expand the indications for gallbladder removal."

Surgical management (cholecystectomy)

Currently, laparoscopic cholecystectomy is the gold standard in the management of cholelithiasis, given the safety and ease of performance and the many well-known advantages of this approach over the conventional open cholecystectomy, including short hospital stay, lesser need for postoperative analgesia, better cosmetic results, fast recovery to full preoperative activity, and avoidance of long-term complications (i.e., incisional hernia) [52, 53]. Conversion to open cholecystectomy may be required for a small percentage of patients when the laparoscopic approach is associated with operative difficulties or when the operating surgeon

Table 2 Expectant management (observation alone)

Advantages
Natural history of asymptomatic gallstones is benign
Only 1–4% of patients per year develop symptoms (biliary "colic")
Only 10 and 20% of patients will develop symptoms 5- and 20-years after the initial diagnosis
Acute biliary symptoms occur in 3%
Almost all patients develop some symptoms (usually biliary colic) before complications arise
Avoidance of a procedure that represents overtreatment for the majority of asymptomatic patients
Avoidance of surgery/anesthesia-related morbidity and mortality
Avoidance of unnecessary cost/workload for the health system
The patient can be operated if and when he or she becomes symptomatic (usually after an episode of biliary colic)
Avoidance of unnecessary surgery/anesthesia in patients with significant comorbidity/advanced age
Disadvantages
Theoretical potential (albeit small) for the development of serious complications of gallstone disease (such as pancreatitis or development of gallbladder cancer)
About 20% of patients will develop symptoms after 20 years
Compared to elective, emergency surgery is associated with increased morbidity and mortality; mean operating time, mean hospitalization and conversion (to open cholecystectomy) rates are higher in emergency surgery
Expected management is contraindicated for selected subgroups of patients with asymptomatic cholelithiasis (see Table 5)

feels at surgery that laparoscopic cholecystectomy cannot be performed safely for a variety of reasons (e.g., presence of firm adhesions owing to previous abdominal operations, difficulty in identifying vital anatomical structures, etc). About 15 years after its introduction, laparoscopic cholecystectomy is considered to be a safe operation, with low morbidity and overall mortality ranging from 0.14–0.50% in different studies, depending on the age and fitness of the patients [54]. However, despite the fact that morbidity has decreased as the years go by, as more operative experience is gained, and experienced in laparoscopy surgeons are involved in the training of the new generation of surgeons [55], complications—potentially severe—do exist and should not be neglected (Table 3).

Routine cholecystectomy

As discussed previously, most authors agree that routine cholecystectomy is not indicated for asymptomatic cholelithiasis. Some, however, maintain that all asymptomatic patients should routinely be operated. The main argument is that surgery can be performed much more safely before the development of potentially serious or even fatal complications (such as acute cholecystitis, cholangitis, pancreatitis), something that usually occurs at a more advanced age and may demand urgent surgery in older patients, with higher morbidity and mortality [5, 40–43, 56]. In contrast, in young, low-risk patients, laparoscopic cholecystectomy is an almost innocuous procedure with low morbidity and practically no mortality [5]. According to this surgical philosophy, precluding or postponing laparoscopic cholecystectomy may be more risky than undertaking an elective operation [5]. Even in the prelaparoscopic cholecystectomy era, Glenn stated that "It is reasonable to strongly recommend an early (open) cholecystectomy for gallstones, whether or not they cause symptoms, unless there is a contraindication to operation. The optimum treatment for asymptomatic cholelithiasis is elective cholecystectomy without undue delay" [40, 41]. Laparoscopic cholecystectomy is easier in asymptomatic than in symptomatic patients, with lesser operative time (92.1 versus 106 minutes), lower conversion rate to open cholecystectomy (1.57% versus 4.6%), and lesser morbidity (4.72% versus 8.80%) ($p < 0.05$) [57]. Although mortality after laparoscopic cholecystectomy is 0.6% for all age groups, it increases with aging, mainly because of the presence of significant comorbidities in older patients (0.14–0.4% in patients <50 years, and 4.5% in those >65 years) [19, 58]. Operative mortality also increases after a "difficult" cholecystectomy (defined as an operation performed after a biliary complication, usually acute cholecystitis or choledocholithiasis, has occurred, even if the acute event has subsided) compared to a "simple" cholecystectomy (prophylactic cholecystectomy or cholecystectomy after an episode

Table 3 Complications of laparoscopic cholecystectomy

Common bile duct injury (1%)
Bile leak (3%)
Bleeding (as in open cholecystectomy)
Retained (undetected) stones and/or bile and stone spillage (with the potential for abscess formation)
Wound infection (1–2%)
Injury to organ/vessels during trocar insertion or during surgery (very rare)
Complications due to pneumoperitoneum (i.e., gas embolism, subcutaneous emphysema) (very rare)
General postoperative complications (ie, deep vein thrombosis, pulmonary embolus, pneumonia, myocardial infarction, atelectasia—pneumonia, stroke) (as in open cholecystectomy)

Table 4 Elective routine laparoscopic cholecystectomy for asymptomatic cholelithiasis

Advantages
Definitive cure of the patient from a disease with a "benign" natural history, but with the potential for the development of serious and even fatal complications (albeit rare) (including the development of gallbladder cancer)
Very safe procedure, with low morbidity and virtually no mortality among low-risk patients. Patient's problem can be relieved by undergoing a safe operation, which becomes even safer for patients with asymptomatic cholelithiasis, owing to favorable local conditions (ie, absence of inflammation, adhesions)
Disadvantages
Overtreatment of a large number of asymptomatic patients who will never develop symptoms or complications
Potential morbidity and even mortality of surgery/anesthesia
Increased cost/workload for the health system

of uncomplicated biliary pain) (mortality, 0.4% for patients with chronic cholecystitis, 1.2% for those with acute cholecystitis, 1.2–1.6% after choledochotomy and common bile duct exploration) [5, 32, 58]. Interestingly, in the study by Morgenstern *et al*. [59] all the deaths after cholecystectomy occurred in patients older than 66 years (mean age, 80.5 years). These authors confirmed also that the mortality increased threefold when choledochotomy become necessary [59]. Table 4 summarizes the advantages and disadvantages of elective (routine) cholecystectomy in patients with asymptomatic cholelithiasis.

Selective cholecystectomy

Laparoscopic cholecystectomy can be performed in selected subgroups of patients with asymptomatic cholelithiasis, who are at greater risk for the development of symptoms or complications (Table 5) [5, 11, 50–52, 60].

Chronic hemolytic syndromes

Patients suffering from chronic hemolytic syndromes (such as sickle cell disease [SCD]) are at risk for gallstone development at a young age (as a result of repeated hemolytic crises). Pigment gallstones are reported in 58% of patients with homozygous SCD and in 17% of patients with heterozygous types of hemoglobinopathies [61, 62]. Two thirds of patients with gallstones have symptoms, although it is often difficult to distinguish between a sickle cell crisis and acute cholecystitis [63]. Patients with other hemolytic anemias are also at risk for gallstone development and many will become symptomatic [64]. Laparoscopic cholecystectomy should be considered for asymptomatic patients with chronic hemolytic syndromes for many reasons [65, 66]:

- Biliary complications of gallstone disease and vaso-occlussive crisis both present with similar manifestations

Table 5 Selective cholecystectomy for asymptomatic cholelithiasis

Clear indications
Suspicion/risk of malignancy
Gallstones associated with gallbladder polyps >1 cm in diameter
Calcified (porcelain) gallbladder
Some ethnic groups or subjects living in areas with high prevalence of gallbladder cancer associated with gallstones (American Indians, Mexican Americans; Colombia, Chile, Bolivia, Maori population of New Zealand)
Presence of large (≥3 cm) gallstones
Asymptomatic cholelithiasis associated with choledocholithiasis
Transplant patients (before or during transplantation)
Chronic hemolytic conditions (sickle cell anemia)
Relative indications
Increased risk of conversion from asymptomatic to symptomatic disease
Life expectancy >20 years
Calculi >2 cm in diameter
Calculi <3 mm and patent cystic duct
Nonfunctioning gallbladder
Diabetes mellitus
Vague dyspeptic symptoms in the presence of gallstones
Questionable indications
Patient living in an area remote from medical facilities
Incidental (concomitant) cholecystectomy during another abdominal operation

(nausea, abdominal pain, fever, leukocytosis, and cholestatic jaundice) and therefore differential diagnosis may be difficult.
- The onset of gallstones at a young age in SCD raises the lifetime risk of biliary complications, and therefore, cholecystectomy following the diagnosis of asymptomatic gallstones in patients with SCD is advisable and justified.

Interestingly, the approach of elective cholecystectomy for asymptomatic cholelithiasis in SCD patients was not generally accepted until recently; in the past, surgery in SCD was associated with a high morbidity and mortality owing to vaso-occlusive crises [61, 67]. However, with the introduction of laparoscopic cholecystectomy, the establishment of its safety in patients with SCD and cholelithiasis, and the significant improvement of anesthetic techniques, more patients (adults and children) are being referred for laparoscopic cholecystectomy [68, 69]. Pediatricians have started screening their patients for gallstones and referring them for laparoscopic cholecystectomy before the development of symptoms and/or complications of the disease [70]. Reduction of Hb-S to a level lower than 50% by preoperative partial exchange transfusion is believed to be associated with a lower risk of veno-occlusive crises [61, 71]. Avoiding open cholecystectomy during acute crises or acute cholecystitis has been advocated [65]. In the laparoscopic era, although surgery should be avoided during veno-occlusive crises, laparoscopic cholecystectomy for acute cholecystitis is not a contraindication provided that all the precautions to guard against a veno-occlusive crisis are taken [3].

Transplantation

Prophylactic cholecystectomy should be strongly considered for patients with asymptomatic gallstones waiting to undergo solid organ transplantation [72, 73]. Prophylactic cholecystectomy can be performed either during the pretransplant period or—when appropriate—at the time of transplantation. The theoretical basis for this recommendation is that these patients are more likely to become symptomatic, especially in the first 2 years after transplantation [73]. Moreover, because of immunosuppression, diagnosis of complications of cholelithiasis may be more difficult; these complications are associated with increased morbidity and mortality. The aim of prophylactic cholecystectomy is to remove of a possible septic focus that carries a high potential for severe complications in immunosuppressed patients [72, 73]. The mortality rate associated with emergency cholecystectomy in patients who have received a heart transplant is high, up to 36% in the review by Begos *et al*. [74]. Finally, cyclosporine and tacrolimus (FK 506), used as immunosuppressive agents, are prolithogenic because of decreased bile salt export pump function [19]. Episodes of acute cholecystitis during episodes of maximum immunosuppression or during rejection episodes have been reported.

Not all authors, however, concur with this aggressive surgical philosophy in transplant patients [75–79]. Greenstein *et al*. [76] follow 21 renal transplant patients with silent gallstones for 4 years. Thirteen patients (87%) remained asymptomatic, and 3 patients (1 had diabetes) developed acute cholecystitis and underwent laparoscopic cholecystectomy with no complications. Similarly, Courcoulas *et al*. [79] followed up 26 transplant patients (including renal, heart, double lung, and heart–lung recipients) who underwent laparoscopic cholecystectomy; 73% of these patients had symptomatic gallstones. These authors found that laparoscopic cholecystectomy is as safe in the transplant population as in the general population and the advantages of short hospital stay, low morbidity and early return to preoperative routines remain equivalent. Melvin *et al*. [78] found that in renal transplant patients, when surgery for gallstones is needed, it was associated with a low risk and does not represent an increased rate of complications in renal transplant patients with 1-, 2-, and 5-year graft survival or 98%, 96%, and 80%, respectively. According to these authors, expectant policy should be exercised in transplant patients with silent gallstones. Once gallstones become symptomatic, laparoscopic cholecystectomy can be safely performed with no adverse effect on morbidity, mortality or graft survival.

Gallbladder carcinoma

Gallbladder cancer, although rare in most Caucasian populations, is amongst the most frequently observed cancers in native populations of North and South America, and in the Maori population of New Zealand, possibly as a result of early onset of gallstones [80–83]. The increased incidence of gallstones (at an early age) in these ethnic groups may be due to the presence of cholesterol lithogenic genes that are highly prevalent in these populations [83]. North American Indian women develop gallbladder carcinoma with a greater frequency than heavy smokers develop pulmonary cancer [5, 84]. In all populations, there is a strong correlation between gallstones and gallbladder cancer. The risk of gallbladder cancer is approximately four times higher in cases with than in those without gallstones. It is estimated that about 80% of patients developing gallbladder carcinoma have gallstones, especially large stones ($\geq$3 cm) [27]. The risk of underlying malignancy is also high in patients with gallbladder polyps larger than 10 mm in diameter [81, 85–87]. Calcified or porcelain gallbladder is associated with carcinoma in 13–25% of patients [8, 88–90]. Recently, some authors questioned the association of porcelain gallbladder with gallbladder carcinoma [91, 92]. According to these authors, a calcified gallbladder is indeed associated with an

increased risk of gallbladder cancer, but at a much lower rate than previously estimated [91]. Interestingly, Stephen *et al.* [91] found that the incidence of cancer depends on the pattern of calcification, with selective mucosal calcification being associated with a greater risk compared to diffuse intramural calcification.

When examining the role of cholecystectomy as a therapeutic strategy to prevent the development of gallbladder carcinoma, the surgeon and the patient should acknowledge that—despite the association of gallstones with gallbladder carcinoma—the risk of developing cancer in all patients with asymptomatic gallstones is less than 0.01%, less than the mortality associated with cholecystectomy [12, 17]. Therefore, prophylactic cholecystectomy is not indicated for the general population with asymptomatic cholelithiasis to prevent future gallbladder cancer [12, 17, 27, 64, 93, 94]. Nevertheless, based on the above data, prophylactic cholecystectomy to prevent gallbladder carcinoma should be strongly considered in selected subgroups of patients with silent gallstones, such as in patients of some ethnic groups living in areas where gallbladder carcinoma is prevalent (American Indians and Mexican Americans, Colombia, Chile, Bolivia) [33, 80–83, 95, 96]. Indications for cholecystectomy should be liberalized in these high-risk populations [5]. The increasing frequency of laparoscopic cholecystectomy in these geographic areas, and the lower threshold for referral, probably will lower the incidence and mortality rates for this lethal disease [81, 97]. Moreover, prophylactic cholecystectomy is indicated in patients with gallbladder polyps larger than 10 mm in diameter and in patients with large gallstones (>3 cm) (see above) [27, 33, 81, 85–87, 97] (Table 5). Finally, and despite some controversy [91, 92], most authors recommend selective prophylactic cholecystectomy for patients with porcelain gallbladder to prevent the development of gallbladder carcinoma [8, 11, 12, 88–90].

Diabetes mellitus

Prophylactic cholecystectomy has been recommended for diabetic patients with silent gallstones [5]. This approach has been based on the belief that diabetic patients belong to the high-risk group for the development of complications of gallstone disease (such as infected bile, gangrenous changes and perforation of the gallbladder), that are more severe than in the general population [98, 99]. Earlier reports found that the risk of acute cholecystitis and subsequent perioperative morbidity and mortality was significantly higher in diabetic compared to nondiabetic patients [5, 100, 101]. It is believed that the autonomic neuropathy in diabetics may mask the pain and other clinical signs associated with acute cholecystitis [102], thereby delaying accurate diagnosis and appropriate management. Therefore, in the past surgeons were urged to consider a diabetic as a high-risk group and prophylactic cholecystectomy was recommended [5, 94, 100].

Recent evidence, however, has shown that the natural history of gallstones in diabetics is generally more benign than thought in the past, with a low risk of a major complications [103]. The cumulative percentage of initially asymptomatic non–insulin-dependent diabetic patients who presented with symptoms and complications was small (14.9% and 4.2%, respectively) [103]. Also, diabetes as an independent risk factor for the formation of gallstones has been questioned [104] and the prevalence of gallstones was found to be similar among diabetic patients (14.4%) and control subjects (12.5%), in a case-control analysis [105, 106]. Moreover, the rates of operative morbidity and mortality for biliary surgery in diabetics currently are comparable, with rates in nondiabetics once other comorbidities such as cardiovascular and renal disease are taken into consideration [98, 107–109]. Therefore, there is no clear benefit to prophylactic cholecystectomy in diabetic patients with asymptomatic gallstones, because surgery does not appear to increase either the duration or quality of life, but may in fact reduce it [11, 32, 98, 103, 107, 109–111]. Consequently, diabetic patients should be managed expectantly with the same criteria as the general population [3, 98, 110, 112]. However, early elective cholecystectomy is advocated once symptoms develop [105].

Vague dyspeptic symptoms

Although patients with gallstones who complain of nonspecific dyspeptic symptoms (such as vague abdominal pain, bloating, fullness, and/or belching) without biliary colic are less likely to improve following cholecystectomy, a large percentage (up to 70%) of them still benefit from surgery [6, 113]. This suggests that indeed these vague, dyspeptic symptoms may be caused by gallstones; therefore, laparoscopic cholecystectomy is expected to improve significantly the quality of life in patients with asymptomatic cholelithiasis who reported vague symptoms [6]. An important practical question, however, is if these patients are truly "asymptomatic" and if gallstones are really "silent."

Other indications for selective cholecystectomy in asymptomatic cholelithiasis

Asymptomatic cholelithiasis in association with stones in the common bile duct is another indication for selective prophylactic surgery, because the presence of ductal stones predispose to potentially severe complications in a significant percentage of patients (up to 50%) [114, 115].

Selective cholecystectomy should also be considered for patients with silent gallstones living in a part of the world very remote from medical facilities. These patients may be

at risk for an adverse outcome should a complication of gallstone disease develop.

Incidental, concomitant cholecystectomy for asymptomatic cholelithiasis during another abdominal operation

Concomitant cholecystectomy for asymptomatic cholelithiasis (diagnosed either preoperatively or intraoperatively) during a planned abdominal operation is a common clinical scenario. Several studies showed a high (up to 70%) incidence of symptoms and/or complications from the biliary system (such as biliary colic, acute cholecystitis, jaundice) in patients with asymptomatic cholelithiasis following laparotomy for unrelated conditions; cholecystectomy was required in a large percentage (up to 40%) of these patients within 1 year of the initial operation [116–121]. The aim of incidental cholecystectomy in this case is to prevent postoperative cholecystitis or the later development of symptoms. Of course the addition of cholecystectomy should not add risks to the patient. In most patients, cholecystectomy "en passant" can be performed safely [119, 121]; therefore, concomitant cholecystectomy during another intraabdominal procedure is a reasonable option for the vast majority of patients [116, 121], unless specific contraindications exist. Ideally, gallstones should be detected preoperatively by ultrasonography; this allows the discussion with the patient before surgery to obtain his or her consent for cholecystectomy. The discussion should emphasize the safety and purpose of the procedure and not dismiss the possible complications, albeit rare, as with any additional surgical procedure. In addition, preoperative diagnosis of silent gallstone disease allows a better planning of the incision [121].

Cholecystectomy-related complications can be avoided by using the proper surgical technique, including proper exposure, by performing an uncomplicated primary operation, and by proper patient selection taking into account comorbidities and general health [122]. Obviously, this strategy is not recommended for high-risk patients, with significant comorbidities, where a minimal operating and anesthesia time is required for an uneventful recovery. Clinical judgment at the time of surgery and caution are required from the part of surgeon to select the optimal management option for the individual patient. Local conditions (eg, presence of a shrunken or scarred gallbladder, cirrhosis, extensive firm adhesions, and/or tissue scarring) should be taken into consideration. Incidental cholecystectomy for asymptomatic cholelithiasis is contraindicated when a prosthetic material (such as vascular graft, mesh for incisional hernia repair) is used during surgery [123]. The performance of concomitant cholecystectomy may be more difficult in the case of a pelvic (gynecologic) procedure, because it may require an additional or extended incision. However, this poses no problem if the pelvic procedure is conducted laparoscopically [3].

Comments

Laparoscopy not only simplified the treatment of cholelithiasis, because of the many and clear advantages of the minimally invasive approach compared to the open method, but also resulted in a broadening of the indications of cholecystectomy and in a decrease of "the surgical threshold" for the surgical management of patients with asymptomatic cholelithiasis. This caused a worldwide increase of the number of cholecystectomies by 18.7% between 1989 and 1993, with an increase of 25% in the age group of 45–64 years [17, 99, 124–125]. This may reflect a change in surgeons' attitude toward some of the indications for cholecystectomy, including asymptomatic gallstones [99, 124]. Surgeons may be treating asymptomatic gallstone disease or resorting to laparoscopic cholecystectomy as a "diagnostic therapeutic test" [127]. This liberal surgical attitude has been further promoted by a lower referral threshold by general practitioners and gastroenterologists asking for surgical treatment by laparoscopic cholecystectomy, as a result of the enhanced perceived benefits of laparoscopic cholecystectomy. These referring physicians often warn patients that they are at an increased risk for the development of severe and potentially lethal complications, causing an unnecessary anxiety to them and suggesting surgery despite the lack of sufficient data to support such an aggressive management approach. Based on existing evidence, physicians should frame their discussions with patients in such a manner as to inform them of the relative risk of an expectant approach versus cholecystectomy, thereby allaying unfounded fears of the expectant management approach. Another significant reason for the increase of the number of (laparoscopic) cholecystectomies is the increased acceptability and demand of minimally invasive surgery by patients, so that few patients refuse surgical treatment, especially after the diagnosis of a "potentially lethal" disease is discussed with the referring physician [124].

Minilaparotomy (open) cholecystectomy is another alternative, but worldwide most surgeons (and patients) prefer the laparoscopic approach. Unfortunately, often both the referring physician and the patient do not take into account the natural history of asymptomatic cholelithiasis and the operation/anesthesia-related potential morbidity and mortality. The end result is the unnecessary overtreatment of a large percentage of subjects with asymptomatic cholelithiasis and the very long lists of patients waiting to undergo cholecystectomy [52]. Among these, of course, are included patients who are symptomatic and who have absolute indication for surgery; the long waiting lists may delay surgery for these symptomatic patients with potential adverse effects

(development of complications, higher conversion rates to open cholecystectomy, increased morbidity/mortality, prolonged hospitalization, etc) [128, 129]. The increased cost and workload for the health system, because of a costly and unnecessary operation not indicated for most patients with asymptomatic cholelithiasis, is another factor that should be taken into consideration when discussing such a controversial issue [130]. Given the high prevalence of gallstones, the cost of prophylactic cholecystectomy is almost four times that of expectant management. It has been estimated that the average cost in Britain is about 5.89 million euros/10,000 patients with asymptomatic cholelithiasis [30]. Considering these numbers and the limited financial resources in health care, and taking into account the natural history of asymptomatic cholelithiasis, it appears unreasonable to treat every patient diagnosed with silent gallstones with cholecystectomy [11].

All patients with asymptomatic cholelithiasis should be fully informed about the natural history of silent gallstone disease and about management options. Existing data support a conservative approach for the large majority of patients with asymptomatic cholelithiasis. Surgery should be offered to selected subgroup of asymptomatic patients with gallstone disease (see Table 5). A careful analysis of hepatobiliary and systemic risk factors (such as advanced age and significant comorbidity) should precede any decision regarding cholecystectomy for asymptomatic cholelithiasis. Patient preferences should be taken into consideration by the surgeon, because the decision for surgery remains an individual decision. Patients unwilling to accept the minimal but real possibility of complications of gallstone disease may prefer to undergo laparoscopic cholecystectomy, but—at the same time—they should acknowledge the morbidity and mortality of the procedure and general anesthesia, albeit minimal. The surgeon should emphasize to the patient that the majority of patients with asymptomatic cholelithiasis rarely develop complications without having at least one episode of biliary colic. Consequently, laparoscopic cholecystectomy can be performed as an elective procedure for the—then symptomatic—cholelithiasis, allowing a more optimal and reasonable timing for surgery. With modern surgical and anesthetic techniques, cholecystectomy—after symptoms develop—results in a mortality rate almost equivalent to elective cholecystectomy.

In conclusion, the evolution of laparoscopy should not alter the indications of cholecystectomy. Because most asymptomatic gallstones remain clinically "silent," routine laparoscopic cholecystectomy is not indicated for the vast majority of subjects with asymptomatic cholelithiasis. Laparoscopic cholecystectomy is a costly procedure performed under general anesthesia and is associated with the potential of morbidity (sometimes serious) and mortality. The risks of the operation outweigh the complications if stones are left in situ. However, laparoscopic cholecystectomy should be performed in selected subgroups of patients with asymptomatic cholelithiasis (see above). Patients should be fully informed about the natural history and available management options, their advantages and disadvantages, and be actively involved in the decision making.

References

1. Tangedahl TN (1987) Therapeutic options for gallstones. Postgrad Med 81:201–203
2. Coleman MJ, Ham JM, Watts JM, *et al*. (1987) A debate: asymptomatic gallstones should not be removed. Aust N Z J Surg 57:897–903
3. Meshikhes AW (2002) Asymptomatic gallstones in the laparoscopic era. J R Coll Surg Edinb 47:742–748
4. Capocaccia L, Giunchi G, Pocchiari F, *et al*. (1984) Prevalence of gallstone disease in an Italian adult female population. Rome Group for the Epidemiology and Prevention of Cholelithiasis (GREPCO). Am J Epidemiol 119:796–805
5. Patino JF, Quintero GA (1998) Asymptomatic cholelithiasis revisited. World J Surg 22:1119–1124
6. Mentes BB, Akin M, Irkorucu O, Tatlicioglou E, Ferahkose Z, Yildirim A, Maral I (2001) Gastrointestinal quality of life in patients with symptomatic or asymptomatic cholelithiasis before and after laparoscopic cholecystectomy. Surg Endosc 15:1267–1272
7. Karam J, Roslyn J (1997). Cholelithiasis and cholecystectomy. In: Zinner MJ, Schwartz SI, Ellis H (eds) Maingot's abdominal operations. Appleton & Lange, Stamford, CT, pp 1723–1725
8. Tait N, Little JM (1995) The treatment of gallstones. BMJ 311:99–105
9. Festi D, Sotitili S, Colecchia A, *et al*. (1999) Clinical manifestations of gallstone disease: evidence from the Multicenter Italian Study on Cholelithiasis (MICOL). Hepatology 30:839–846
10. NIH Consensus Statement (1992) Gallstones and laparoscopic cholecystectomy (Vol. 10, No. 3, September 14–16, 1992). National Institutes of Health, U.S. Department of Health and Human Services, Office of Medical Applications of Research, Bethesda, MD
11. Vauthey JN, Saldinger PF (2000) The natural history of gallstones and asymptomatic gallstones. In: Blumgart LH, Fong Y (eds) Surgery of the liver and biliary tract, 3rd edn, Elsevier, New York, pp 643–649.
12. Allen J, Cushieri A (2005) Cholelithiasis. http://www.emedicine.com/med/topic836.htm. Updated April 26, 2005
13. The Rome Group for Epidemiology and Prevention of Cholelithiasis (GREPCO) (1988) The epidemiology of gallstone disease in Rome, Italy. Part I. Prevalence data in men. Hepatology 8:904–906
14. The Rome Group for Epidemiology and Prevention of Cholelithiasis (GREPCO) (1988) The epidemiology of gallstone disease in Rome, Italy. Part II. Factors associated with the disease. Hepatology 8:907–913
15. Attili AF, Carulli N, Roda E, *et al*. (1995) Epidemiology of gallstone disease in Italy: prevalence data of the Multicenter Italian Study on Cholelithiasis (M.I.COL.). Am J Epidemiol 141:158–165
16. Muhrbeck O, Ahlberg J (1995) Prevalence of gallstone disease in a Swedish population. Scand J Gastroenterol 30:1125–1128
17. Beckingham IJ (2001) ABC of diseases of liver, pancreas, and biliary system: Gallstone disease. BMJ 322:91–94

18. Misciagna G, Leoci C, Guerra V, *et al*. (1996) Epidemiology of cholelithiasis in southern Italy. Part II: risk factors. Eur J Gastroenterol Hepatol 8:585–593
19. Johnson AG, Fried M, Tytgat GNJ, Krabshuis JH. OMGE Position Statement. Asymptomatic gallstone disease. http://www.omge.org/guidelines/statement04/s_data4_en.htm
20. Ransohoff DF, Gracie WA (1993) Treatment of gallstones. Ann Intern Med 119:606–619
21. Potts JR 3rd (1990) What are the indications for cholecystectomy? Cleve Clin J Med 57:40–47
22. Fendrick AM, Gleeson SP, Cabana MD, *et al*. (1993) Asymptomatic gallstones revisited. Is there a role for laparoscopic cholecystectomy? Arch Fam Med 2:959–968
23. Marshall JB (1994) Current opinions in gallstone management. What to do when symptoms are mild or absent? Postgrad Med 95:115–121
24. Persson GE (1996) Expectant management of patients with gallbladder stones diagnosed at planned investigation. A prospective five-seven year follow-up study of 153 patients. Scand J Gastroenterol 31:191–199
25. Friedman GD (1993) Natural history of asymptomatic and symptomatic gallstones. Am J Surg 165:399–404
26. Cucchiaro G, Watters CR, Rossitch JC, *et al*. (1989) Deaths from gallstones. Incidence and associated clinical factors. Ann Surg 209:149–151
27. Godrey PJ (1984) Gallstones and mortality: a study of all gallstone related deaths in a single health district. Gut 25:1029–1033
28. Wetter LA, Way LW (1991) Surgical therapy for gallstone disease. Gastroenterol Clin North Am 20:157–169
29. Zubler J, Markowski G, Yale S, Graham R, Rosenthal T (1998) Natural history of asymptomatic cholelithiasis in family practice office practices. Arch Fam Med 7:230–233
30. Fazili FM. To operate or not to operate on asymptomatic gallstones in the laparoscopy era. www.laparoscopyhospital.com/NEWS.HTM
31. Attili AF, De Santis A, Capri R, *et al*. (1995) The natural history of gallstones: the GREPCO experience. The GREPCO group. Hepatology 21:655–660
32. Gracie WA, Ransohoff (1982) The natural history of gallstones: the innocent gallstone is not a myth. New Engl J Med 309:78–80
33. Lowenfels AB, Domellof L, Lindstrom CG, *et al*. (1982) Cholelithiasis, cholecystectomy, and cancer; a case-control study in Sweden. Gastroenterology 83:672–676
34. Hermann RE (1989) The spectrum of biliary stone disease. Am J Surg 158:171–173
35. Wada K, Imamura T (1993) Natural course of asymptomatic gallstone disease. Nippon Rinsho 51:1737–1743
36. McSherry CK, Ferstenberg H, Calhoun WF, *et al*. (1985) The natural history of diagnosed gallstone disease in symptomatic and asymptomatic patients. Ann Surg 202:59–63
37. Cucchiaro G, Rossitch JC, Bowie J, *et al*. (1990) Clinical significance of ultrasonographically detected coincidental gallstones. Dig Dis Sci 35:417–421
38. Friedman GD, Raviola CA, Fireman B (1989) Prognosis of gallstones with mild or no symptoms: 25 years of follow-up in a health maintenance organization. J Clin Epidemiol 42:127–131
39. Wacha H, Ungeheuer E (1986) The asymptomatic gallstone. Langenbecks Arch Chir 369:83–87
40. Glenn F (1981) Surgical management of acute cholecystitis in patients over 65 years of age and older. Ann Surg 193:56–61
41. Glenn F (1981) Silent gallstones. Ann Surg 193:251–253
42. Gracie WA, Ransohoff DR (1983) The silent stone requiescat in pace. In: Delaney JP, Varco RL (eds) Controversies in surgery II. Saunders, Philadelphia, pp 361–370
43. McSherry CK, Glenn F (1980) The incidence and causes of death following surgery for non-malignant biliary tract disease. Ann Surg 191:271–275
44. Ransohoff DG, Gracie WA, Wolfenson LB, *et al*. (1983) Prophylactic cholecystectomy or expectant management for silent gallstones. Ann Intern Med 99:199–204
45. Thistle JL, Cleary PA, Lachin JM, *et al*. (1984) The natural history of cholelithiasis: the National Cooperative Gallstone Study. Ann Intern Med 101:171–177
46. Houssin D, Castaing D, Lemoine J, Bismuth H (1983) Microlithiasis of the gallbladder. Surg Gynecol Obstet 157:20–23
47. Lee SH, Burhenne HJ (1990) Gallbladder surgery following cholecyst-lithotripsy: suggested guidelines for treatment. Br J Surg 77:1268–1273
48. Lee SP, Nichols JF, Park HZ (1992) Biliary sludge as a cause of acute pancreatitis. N Engl J Med 326:589–593
49. American College of Physicians (1993) Guidelines for the treatment of gallstones. Ann Intern Med 119:620–622
50. Mulvihill SJ (2003) Surgical management of gallstone disease and postoperative complications. Semin Gastrointest Dis 14:237–244
51. Patient Care Committee, Society for Surgery of the Alimentary Tract (SSAT) (2004) Treatment of gallstone and gallbladder disease. SSAT patient care guidelines. J Gastrointest Surg 8:363–364
52. Keullemans YCA, Venneman NG, Gouma DJ, van Berge Henegouwen GP (2002) New strategies for the treatment of gallstone disease. Scand J Gastroenterol 37(Suppl 236):87–90
53. Svanvik J (2000) Results of laparoscopic compared with open cholecystectomy. Eur J Surg Suppl 585:12–15
54. Maxwell JG, Tyler BA, Rutledge R, Brinker CC, Maxwell BG, Covington DL (1998) Cholecystectomy in patients aged 80 and older. Am J Surg 176:627–631
55. Gadacz TR (2000) Update on laparoscopic cholecystectomy, including a clinical pathway. Surg Clin North Am 80:1127–1145
56. Lund J (1960) Surgical indications in cholelithiasis: prophylactic cholecystectomy elucidated on the basis of long-term follow-up in 526 non-operated cases. Ann Surg 151:153–158
57. Yano H, Kinuta M, Iwazana T, Kanoh T, Monden T (2003) Laparoscopic cholecystectomy for asymptomatic cholelithiasis. Dig End 15:190–195
58. McSherry CK (1989) Cholecystectomy: the gold standard. Am J Surg 158:174–178
59. Morgenstern L, Wong L, Berci G (1992) Twelve hundred open cholecystectomies before the laparoscopic era. A standard for comparison. Arch Surg 127:400–403
60. Mackie R (1993) Cholelithiasis; the alternatives to surgery. In: Graber JN, Schultz LS, Pietrafitta JJ, Hichock DF (eds) Laparoscopic abdominal surgery, McGraw-Hill Inc, New York, pp 203–216
61. Bond LR, Hatty SR, Horn ME, *et al*. (1987) Gallstones in sickle cell disease in the United Kingdom. BMJ 295:234–236
62. Heeney MM (2005) Risk of cholelithiasis in sickle cell anemia. Haematologica 90:147a
63. Serafini AN, Spoliansky G, Sfakianakis GN, *et al*. (1987) Diagnostic studies in patients with sickle cell anemia and acute abdominal pain. Arch Intern Med 147:1061–1062
64. Goldfarb A, Grisaru D, Gimmon Z, *et al*. (1990) High incidence of cholelithiasis in older patients with homozygous beta-thalassemia. Acta Hematol 83:120–122
65. Stephens CG, Scott RB (1980) Cholelithiasis in sickle cell anemia: surgical or medical management. Arch Intern Med 140:648–651

66. Ware R, Filston HC, Schultz WH, Kinney TR (1988) Elective cholecystectomy in children with sickle hemoglobinopathies. Successful outcome using a preoperative transfusion regimen. Ann Surg 208:17–22
67. Gibson TJ, O'Dell RF, Cathcart RS, 3rd, *et al.* (1979) Treatment of cholelithiasis in patients with sickle cell anemia. South Med J 72:391–392
68. Fall B, Sagna A, Diop PS, *et al.* (2003) Laparoscopic cholecystectomy in sickle cell anemia. Ann Chir 128:702–705
69. Vecchio R, Cacciola E, Di Martino M, *et al.* (2002) Laparoscopic surgery in sickle cell disease. Surg Endosc 16:1807–1808
70. Suell MN, Horton TM, Dishop MK, *et al.* (2004) Outcomes for children with gallbladder abnormalities and sickle cell disease. J Pediatr 145:617–621
71. Al-Mulhim AS, Al-Mulhim FM, Al-Suwaiyagh AA (2002) The role of laparoscopic cholecystectomy in the management of acute cholecystitis in patients with sickle cell disease. Am J Surg 183:668–672
72. Kao LS, Kuhr CS, Flum DR (2003) Should cholecystectomy be performed for asymptomatic cholelithiasis in transplant patients? J Am Coll Surg 197:302–312
73. Gupta D, Sakorafas GH, McGregor CG, Harmsen WS, Farnell MB (2000) Management of biliary tract disease in heart and lung transplant patients. Surgery 128:641–649
74. Begos DG, Franco KL, Baldwin JC, *et al.* (1995) Optimal timing and indications for cholecystectomy in cardiac transplant patients. World J Surg 19:661–665
75. Jackson T, Treleaven D, Arlen D, *et al.* (2005) Management of asymptomatic cholelithiasis for patients awaiting renal transplantation. Surg Endosc 19:510–513
76. Greenstein SM, Katz S, Sun S, *et al.* (1997) Prevalence of asymptomatic cholelithiasis and risk of acute cholecystitis after kidney transplantation. Transplantation 63:1030–1032
77. Lord RV, Ho S, Coleman MJ, *et al.* (1988) Cholecystectomy in cardiothoracic organ transplant recipients. Arch Surg 133:73–79
78. Melvin WS, Meier DJ, Elkhammas EA, *et al.* (1998) Prophylactic cholecystectomy is not indicated following renal transplantation. Am J Surg 175:317–319
79. Courcoulas AP, Kelly E, Harbrecht BG (1996) Laparoscopic cholecystectomy in the transplant population. Surg Endosc 10:516–519
80. Pandey M (2003) Risk factors for gallbladder cancer: a reappraisal. Eur J Cancer Prev 12:15–24
81. Lowenfels AB, Walker AM, Althaus DP, *et al.* (1989) Gallstone growth, size, and risk of gallbladder cancer: an interracial study. Int J Epidemiol 18:50–54
82. Strom BL, Soloway RD, Rios-Dalenz JL, *et al.* (1995) Risk factors for gallbladder cancer. An international collaborative case-control study. Cancer 76:1747–1756
83. Miquel JF, Covarrubias C, Villaroel L, *et al.* (1998) Genetic epidemiology of cholesterol cholelithiasis among Chilean Hispanics, American-Indians, and Maoris. Gastroenterology 115:937–946
84. Paltoo DN, Chu KC (2004) Patterns in cancer incidence among American Indians/Alaska Natives, United States, 1992–1999. Public Health Rep 119:443–451
85. Piehler JM, Crichlow RW (1978) Primary carcinoma of the gallbladder. Surg Gynecol Obstet 147:929–942
86. Chattopadhyay D, Lochan R, Balupuri S, *et al.* (2005) Outcome of gallbladder polypoidal lesions detected by transabdominal ultrasound scanning: a nine year experience. World J Gastroenterol 11:2171–2173
87. Lee KF, Wong J, Li JC, Lai PB (2004) Polypoid lesions of the gallbladder. Am J Surg 188:186–190
88. Ashur H, Siegal B, Oland Y, *et al.* (1978) Calcified gallbladder (porcelain gallbladder). Arch Surg 113:594–596
89. Berk RN, Armbuster TG, Saltzstein SL (1973) Carcinoma in the porcelain gallbladder. Radiology 106:29–31
90. Shimizu M, Miura J, Tanaka T, *et al.* (1989) Porcelain gallbladder: relation between its type by ultrasound and incidence of cancer. J Clin Gastroenterol 11:471–476
91. Stephen AE, Berger DL (2001) Carcinoma in the porcelain gallbladder: a relationship revisited. Surgery 129:699–703
92. Towfigh S, McFadden DW, Cortina GR, *et al.* (2001) Porcelain gallbladder is not associated with gallbladder carcinoma. Am Surg 67:7–10
93. Diehl AK (1983) Gallstone size and the risk of gallbladder cancer. JAMA 250:2323–2329
94. Gibney EJ (1990) Asymptomatic gallstones. Br J Surg 77:368–372
95. Sheth S, Bedford A, Chopra S (2000) Primary gallbladder cancer: recognition of risk factors and the role of prophylactic cholecystectomy. Am J Gastroenterol 95:1402–1410
96. Gupta SK, Shukla VK (2004) Silent gallstones: a therapeutic dilemma. Trop Gastroenterol 25:65–68
97. Lowenfels AB, Maisonneuve P, Boyle P, Zatonski WA (1999) Epidemiology of gallbladder cancer. Hepatogastroenterology 46:1529–1532
98. Aucott JN, Cooper GS, Bloom AD, Aron DC (1993) Management of gallstones in diabetic patients. Arch Intern Med 153:1053–1058
99. Schwesinger WH, Diehl AK (1996) Changing indications for laparoscopic cholecystectomy. Stones without symptoms and symptoms without stones. Surg Clin North Am 76:493–504
100. Ikard RW (1990) Gallstones, cholecystitis, and diabetes. Surg Gynecol Obstet 171:528–532
101. Landau O, Deutsch AA, Kott I, *et al.* (1992) The risk of cholecystectomy for acute cholecystitis in diabetic patients. Hepatogastroenterology 39:437–438
102. Ponte E, Pinebianco A, Morena S (1990) Diabetic neuropathy. Minerva Med 81:335–340
103. DelFavero G, Caroli A, Meggiato T, *et al.* (1994) Natural history of gallstones in non-insulin-dependent diabetes mellitus. A prospective 5-year follow-up. Dig Dis Sci 39:1704–1707
104. Araujo LM, DeOlivera DA, Nunes DS (1998) Liver and biliary ultrasonography in diabetic and non-diabetic obese women. Diabetes Metab 24:458–462
105. Persson GE, Thulin AJ (1991) Prevalence of gallstone disease in patients with diabetes mellitus. A case-control study. Eur J Surg 157:579–582
106. Chapman BA, Wilson IR, Frampton CM, *et al.* (1996) Prevalence of gallbladder disease in diabetes mellitus. Dig Dis Sci 41:2222–2228
107. Sandler RS, Maule WF, Baltus ME (1986) Factors associated with postoperative complications in diabetics after biliary surgery. Gastroenterology 91:157–162
108. Ransohoff DF, Miller GL, Forsythe SB, Hermann RE (1987) Outcome of acute cholecystitis in patients with diabetes mellitus. Ann Intern Med 106:829–832
109. Babineau TJ, Booth A Jr (1995) General surgery consideration in the diabetic patient. Infect Dis Clin North Am 9:183–193
110. Friedman LS, Roberts MS, Brett AS, *et al.* (1988) Management of asymptomatic gallstones in the diabetic patient. A decision analysis. Ann Intern Med 109:913–919
111. Ahmed A, Cheung RC, Keefe EB (2000) Management of gallstones and their complications. Am Pham Phys 61:1673–1678
112. Angelico F, Del Ben M, Barbato A, *et al.* (1997) Ten-year incidence and natural history of gallstone disease in a rural population of women in central Italy. The Rome Group for the Epidemiology and Prevention of Cholelithiasis[GREPCO]. Ital J Gastroenterol Hepatol 29:249–254

113. Karam J, Roslyn J (1997) Cholelithiasis and cholecystectomy. In: Zinner MJ, Schartz SI, Ellis H (eds) Maingot's abdominal operations. Appleton & Lange, Stamford, CT, pp 1723–1725
114. Bateson MC (1984) Gallbladder disease and cholecystectomy rate are independently variable. Lancet 2:621–622
115. Bateson MC, Bouchier IAD (1975) Prevalence of gallstones in Dundee: necropsy study. Br Med J 4:4271–4275
116. Saade C, Bernard D, Morgan S, *et al*. (1987) Should cholecystectomy be done en passant for asymptomatic cholelithiasis? Can J Surg 30:350–353
117. Ouriel K, Ricotta JJ, Adams JT, Deweese JA (1983) Management of cholelithiasis in patients with abdominal aortic aneurysm. Ann Surg 198:717–719
118. String ST (1984) Cholelithiasis and aortic reconstruction. J Vasc Surg 1:664–669
119. Thompson JS, Philben VJ, Hodgson PE (1984) Operative management of incidental cholelithiasis. Am J Surg 148:821–824
120. Bragg LE, Thompson JS (1989) Concomitant cholecystectomy for asymptomatic cholelithiasis. Arch Surg 124:460–462
121. Juhasz ES, Wolff BG, Meagher AP, *et al*. (1994) Incidental cholecystectomy during colorectal surgery. Ann Surg 219:467–472
122. Wolff BG (1995) Current status of incidental surgery. Dis Colon Rect 38:435–441
123. Philosophe BP, Carter S, Doherty GM, *et al*. (1997) Hepatobiliary disease. In: Washington (ed) The Washington manual of surgery. Little Brown, Boston, pp 258–265
124. Lam CM, Murray FE, Cuschieri A (1996) Increased cholecystectomy rate after the introduction of laparoscopic cholecystectomy in Scotland. Gut 38:282–284
126. Legorreta AP, Silber JH, Constantino GN, *et al*. (1993) Increased cholecystectomy rate after the introduction of laparoscopic cholecystectomy. JAMA 270:1429–1432
127. Orlando R, Russell JC, Lynch J, Mattie A (1993) Laparoscopic cholecystectomy. A statewide experience. Arch Surg 128:494–498
128. Spiro HM (1992) Diagnostic laparoscopic cholecystectomy. Lancet 339:167–168
129. Guirguis S, Taylor E (1995) The complication of cholelithiasis is caused by state authorization delays. Surg Endosc 9:974–976
130. Rutledge D, Jones D, Rege R (2000) Consequences of delay in surgical treatment of biliary disease. Am J Surg 180:466–469
131. Fullarton GM, Darling K, Williams J, *et al*. (1994) Evaluation of the cost of laparoscopic and open cholecystectomy. Br J Surg 81:124–126